CISC - Conversation with LLM for Part 2 Final Project Submission.

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can you also make sure to add documentation. also i want a decent amount of possible swim sets in the library

Absolutely! Based on both parts of your project proposal and the final project requirements, I would build this as a complete swim workout generator called SwimSet.

I want to make sure the code accomplishes two things:

1. Meets all of your professor's required programming specifications.
2. Creates something that actually feels like a useful website for swimmers, rather than just a basic assignment.

I've also included a larger workout library, documentation, and comments explaining how the different pieces work.

One important thing: Your assignment specifically says that you need to understand the code and document any LLM assistance. Use this as a working version, read through it, and make sure you can explain the functions before submitting it.

1. Website features

Here is what I would include.

Page	Functionality
Home	Introduces SwimSet and provides navigation
Workout Generator	Allows users to select workout preferences
Generated Workout	Displays the customized workout

Workout Library	Contains 20 different prewritten swim sets
Workout Details	Displays an individual library workout
Favorites	Displays workouts saved by the user
About	Explains how the application works

User inputs

Your three required inputs will be:

- Skill level: Beginner, Intermediate, Advanced
- Workout focus: Endurance, Sprint, Pace Training, Technique, Recovery
- Distance goal: Number of yards

Workout library

I would start with 20 different workouts covering different skill levels and training focuses.

This gives your generator enough variety without making the programming logic unnecessarily complicated.

2. Complete Python code

The code below uses the Drafter library and Python dataclasses.

It includes:

- 2 dataclasses
- 11 routes
- 7 different website pages
- 3 user inputs
- Multiple meaningful conditional statements
- Loops that iterate through lists of dataclass instances
- 20 prewritten swim workouts
- A favorites feature

- Documentation and docstrings
- Basic unit tests

Save this as `swimset.py`.

```
from drafter import *

    Tests whether the workout generator can find

    a workout matching the selected preferences.

    """



    test_state = State(

        "Beginner",

        "Endurance",

        1200,
```

```
    None,

    [],

    create_workout_library()

)




workout = find_workout(test_state)




assert workout is not None

assert workout.skill_level == "Beginner"

assert workout.workout_focus == "Endurance"
```

```
def test_save_favorite():

    """

    Tests whether a generated workout can be added

    to the favorites list.

    """

    library = create_workout_library()

    test_state = State(

        "Beginner",

        "Endurance",

        1200,
```

```
        library[0],


        [],


        library


    )




    save_favorite(test_state)




    assert len(test_state.favorite_workouts) == 1









# Run tests before starting the website.


test_find_workout()
```

```
test_save_favorite()









# ------------------------------------------------


# START WEBSITE


# ------------------------------------------------




start_server(


   State(


       "Beginner",


       "Endurance",


       1500,
```

```
        None,


        [],


        create_workout_library()


    ),


    site_title="SwimSet"


)
```

---

3. How this meets your professor's requirements

This is particularly important because your professor states that missing more than two requirements results in no credit.

Requirement	Where it is implemented

At least 7 routes	11 routes
At least 5 pages	7 pages
State dataclass	`State`
Second dataclass	`SwimSet`
At least 4 State attributes	6 attributes
List of dataclass instances	`favorite_workouts` and `workout_library`

4 meaningfully modified State fields	Skill level, focus, distance, current workout, favorites
At least 3 user inputs	Two SelectBoxes and one TextBox
At least 3 if statements	Workout generation, validation, favorites
At least 1 meaningful loop	Workout library and favorites
Legitimate purpose	Generates swimming workouts
No global variables	No global variables used

Drafter library	Imported and used throughout

One particularly important requirement: Your professor requires four State fields to be meaningfully modified.

This code modifies:

- `skill_level` when the user submits their preferences.
- `workout_focus` when the user submits their preferences.
- `distance_goal` when the user submits their preferences.
- `current_workout` when a workout is generated.
- `favorite_workouts` when a workout is saved.

That gives you five fields that are meaningfully modified.

4. Documentation

You specifically mentioned wanting documentation. I would make this a priority because your professor explicitly requires good docstrings and documentation of advanced coding techniques.

The code already includes:

- A docstring for each function.
- Explanations of both dataclasses.
- Comments separating the different routes.
- Explanations of how the workout library works.
- Explanations of how the loops work.
- Basic unit tests.

I would also recommend reviewing these official resources:

1. Drafter State Documentation — Explains how your State dataclass works.
2. Drafter Forms and Input — Explains how user inputs connect to route parameters.
3. Drafter Components — Explains buttons, lists, headers, and other website components.
4. Drafter Unit Testing — Shows how to test routes and functions.

5. A few things I would improve as you develop it

I intentionally kept the programming relatively straightforward so that it stays appropriate for an introductory computer science project.

However, there are three things worth considering as you work through it.

1. Distance matching

Currently, the generator chooses the workout with the closest distance to the user's goal. It does not construct a completely new workout that exactly matches the requested distance.

For example, if someone requests 2,000 yards, the program might give them a 1,900-yard workout.

That's a reasonable first implementation, but eventually you could improve it.

2. More workout variety

The library currently contains 20 workouts. You could eventually expand this to 30–40, particularly if you want swimmers to be able to generate different workouts without seeing the same ones repeatedly.

3. Favorites

The favorites feature currently stores workouts while the website is open. Because Drafter's State is temporary, reloading the website resets favorites. That's okay for your initial project; permanent storage would add complexity.

My recommendation

I would not try to add more complicated features immediately.

Get the website running first. Then work through the generator, library, and favorites separately. Make sure you understand how the `State` and `SwimSet` dataclasses interact.

Also, since your assignment requires you to include your LLM conversation in Part 7, make sure you document this assistance.

One next step I'd recommend: I can help you test each route against your professor's rubric and fix any Drafter errors before you deploy SwimSet.

Here is part 1 of my final project:Final Project: Pick a ProjectComputer Science - Week 10
Oct 3, 2026 - Version 0.0.2Please make sure to your UD email AND name is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! Every person should submit their own individual worksheet.Talk about each of the questions with a friend or our LLM (or both), but make sure the words you enter below are yours (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.RoleNameEmailDeveloper (You)Natalie Campbellnatcamp@udel.eduFriend (optional)Elena Sangeminoesang@udel.eduPart 1: The Goal & Prior ImplementationsIdentify the Goal: What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major. Biomedical Engineering majors may consider one of the following: If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.Otherwise, create a website related to your BMEG100 Lab Report 1, Cytoskeletal and Nuclear Staining Lab Report Guidelines, linked here (University of Delaware account required to access).Chemistry and Chemical Engineering majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas

into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.Majors in the Electric & Computer Engineering department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images. Mechanical Engineering majors may consider a project that somehow uses both the NumPy and SciPy Python libraries. My goal is to create a web application that will generate a swim set for a workoutPrior Implementations: Search the internet. What existing tools or websites currently attempt to address it?There are some websites or apps that already help swimmers create workouts, like MySwimPro, but most of these require subscriptions or go beyond a casual swim level. Your Implementation: Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?My web application will allow swimmers to enter information such as skill level, workout length, and training focus. The program will then generate a swim set that matches the user's goals. The website is supposed to help swimmers create structured workouts without having to spend time planning them.Part 2: The Input-Process-Output (IPO) ModelThink about the fundamental flow of your program.The Core Goal: Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.The goal of my program is to take user preferences as input, process them using predefined swim sets and rules, and generate a customized swim workout.Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).Skill level (beginner, intermediate, advanced)Practice lengthTotal distance goalWorkout focus(maybe previous times)Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).The program will compare the user's input to a collection of workout sets and pick appropriate warmups, main sets, and cooldowns. It will then organize the set into a complete workout.Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).A formatted swim workout with a warmup, main set, and cooldown.Part 3: Data AbstractionIdentify the "nouns" of your problem.Representing Reality: Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., If building a recipe app, the nouns are "Recipe" and "Ingredient").SwimmerWorkoutSwim SetExerciseIdentifying Properties: For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?Details I need: (e.g.,

Ingredient name, weight, cost)Details I will ignore: (e.g., Ingredient color, brand, expiration date)Need:
- workout type- total distance- difficulty level- team
Ignore:- pool location- time of dayPart 4: Challenges, AI Strategy, & GoalsDifficulties: What parts of this project seem the most difficult or technically complex?The most challenging part will be creating a system that generates workouts that are balanced and appropriate for different skill levels. Another challenge will be making sure that the generated distances match the user's requested workout length.Coding Strategy: If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project. I still haven't decided if I will be using an LLM but if I do I will make sure I read through the suggested code carefully, test it, and understand each function. I will compare suggestions with course materials and debug errors myself before asking for additional help.Goals: What is your ultimate goal for this project? (e.g., Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply).My ultimate goal is to build a website that swimmers could realistically use to generate workouts. I want to improve my programming skills and become more comfortable using Drafter.Part 5: Research & Brainstorming ConversationHave a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.The Initial Prompt/Question: Ask the AI or your friend to help you brainstorm. (e.g., "I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"). Paste your prompt below:"I think I want to create a web tool that could help generate swim workouts. Coming up with a workout set is the most annoying thing I have to do if I am trying to swim alone"The Response: Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?She said that she likes my idea. She also suggested that it could be helpful to add an input where the swimmer can put in previous times and have the program identify some strengths and weaknesses.The Pushback: You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. Paste your response here:The most complex suggestion that my friend gave was having the program be able to store previous workouts and then adjust future workouts based on improvements. An alternative to having it store it was just to maybe add an extra input to allow spots for improvements so the program knows to avoid that specific part.Reflection: Based on this conversation, how has your project idea changed or become more focused?After

brainstorming, I narrowed my project to generating workouts based only on a few user inputs rather than building a complete training platform. This makes it more realistic and achievable.Part 6: Resource PlanningWhat resources will you use if you get stuck? I will use the textbook, lecture materials, office hours, TA's and maybe an LLM for clarifications.If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?Read the full error message carefullyIdentify which function or file caused the errorTest smaller sections of code separatelyCompare my code with examples from classCheck drafter and bakery documentationUse print statements to locate the problemAsk LLM to explainVisit office hours or ask a TA
Part 7: LLM ConversationIf you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.I did not use an LLM for this assignment, I only used a partner to bounce ideas off of.Part 8: SubmitIn Google docs, with this document open, please click on File > Download > Microsoft Word (.docx), and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project. When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.here is part 2:Final Project Design PhaseComputer Science - Week 11
Oct 3, 2026 - Version 0.0.4For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.0) Project Planning and PreparationSuccessful programmers spend time planning before they begin coding. After reviewing the project requirements, answer the following questions to help guide your design and development process. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?This project will use concepts like

functions, routes, conditions, lists, dataclasses, and state variables. I expect to use lists to store swim sets, conditionals to generate different workouts based on user input and loops to display workout information. I want to improve my understanding of state variables while I become more comfortable building a larger project.Which parts of the project do you expect to be the easiest for you? The hardest? Why?I think the hardest part will be transforming my swimming logic into the computer so that it generates workouts that are appropriate for different skill levels while also keeping the total workout distance wanted in mind. The easiest part will most likely be creating the pages and organizing the information since I have a good understanding of swim workouts.Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first? The workout generation logic will likely require the most time because it involves picking workouts based on multiple user inputs/preferences. I think my first step will be creating the dataclasses and state variables. Then I can build the basic pages and navigation. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements? My biggest distraction will be other coursework from classes. To minimize distractions, I plan to work on the project in small sections, a little bit each day, rather than waiting until the deadline. Some resources I plan on using are the textbook, notes, Drafter documentation and office hours. One thing I think could be helpful is seeing a previous example from a student.1) Site DiagramCreate a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.We recommend drawing each

page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page. 2) Implementing RequirementsThis table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.RequirementImplementationPurpose/FunctionalityA 1-2 sentence description of the purpose or functionality of your siteThe website will help swimmers quickly create structured workouts without spending time planning practices themselves. It generates a warmup, main set and cooldown based on the user's preferences.State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass.Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.skill_level: str → updated when the user selects a skill lever (beginner, intermediate, pro)workout_focus: str → updated when the user selects a training focus (endurance, pace training, sprint, technique/drill, recovery)distance_goal: int → updates when the user enters their desired workout distancecurrent_workout: Workout → updates when all inputs are in and and a new workout is generated for the userfavorite_workouts: list[Workout] → updates when the user saves a workout to favoritesSecond DataclassYour second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)Workout- represents individual workout components that can be combined to create a complete practicewarmup: list[str] → list of the warmup setsmain_set: list[str] → list of the main sets of the workoutcooldown: list[str] → list of

cooldown swims for the swimmer to completetotal_distance: int → distance of setRoutes (minimum 7) The names of your route functionshome → main landing pageworkout_generator → page where user inputs informationgenerate_workout → creates the workout based on inputsview_workout → displays generated workout to userworkout_library → displays all available workout ideas and combinations to userfavorites→ displayed saved workoutssave_favorite → saves workout to favoritesabout → displays info about the website *** (might not need this)Pages (minimum 5)A brief description of the pages in your siteHome: main page with navigation linkedWorkout Generator: where the user enters their workout preferencesGenerated Workout: shows warmup, main set, cooldown, and total distanced based on user preferencesWorkout Library: shows example swim sets and workout ideasFavorites: shows workouts saved by the userAbout: explains the purpose of the website and how it works ***User Input (minimum 3)Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposesFor more information on these input types, see Drafter Documentation.SelectBox: Skill Level → user chooses between beginner, intermediate, or advancedSelectBox: Workout Focus → user chooses between endurance, sprint, pace training, technique, or recoveryTextBox: Distance Goal → user enters desired workout distance***CheckBox: Competitive Swimmer (might not keep this because if I did, it would probably involve times)If Statements (minimum 3)Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the siteif skill_level == “beginner”Will generate shorter and less challenging workoutsChanges for each inputif workout_focus == “sprint”Will generate sprint-focused sets with shorter repeatsEtc if different inputif workout_focus == “endurance”Generates longer aerobic sets, etcif distance_goal >2500Adds more sets to increase workout volumeLoop (minimum 1)This loop must iterate over the list/dict of dataclass instances in your State attribute.Description of this loop and why it is meaningful to the siteI am still

workshopping a loop but:A loop will iterate through the list of workouts saved in favorite_workouts. It will display each saved workout on the favorites page, including the warmup, main set, cooldown, and total distance. It is meaningful because it will allow the users to view all of their saved workouts.LLM ConversationIf you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.what is the difference between my state dataclass and my second dataclass and how do i implement my second dataclass into my state oneThe main difference is that your State dataclass and your second dataclass serve two different purposes:State dataclass: Holds the overall information your application needs to run. It acts as the central container for your app.Second dataclass: Represents a specific object or type of information your application needs to keep track of.For example, imagine you are building a study tracker:@dataclassclass StudySession: subject: str hours: float date: strThis second dataclass represents one individual study session.Your State dataclass could then contain a list of those sessions:@dataclassclass State: sessions: list[StudySession]Now your State keeps track of all the StudySession objects.How they work togetherWhen a user creates a new study session, you add it to the list inside State:def add_session(state, subject, hours, date): new_session = StudySession(subject, hours, date) state.sessions.append(new_session) return stateThis way:StudySession defines what an individual session looks like.State stores all the sessions.Your functions can add, remove, or analyze sessions through State.Let's look at your actual codeI can explain exactly how to connect your two dataclasses, but I need to see them first.Paste your State dataclass and your second dataclass here.I'll explain:What each variable in both dataclasses does.Why you need two separate dataclasses.Which variable you should

add to State.How to initialize and use your second dataclass in Drafter.I'll keep the explanation at a CS1 level so you understand how it works rather than just copying code.well im creating both and im not sure how to put them together. basically the goal of my project is to create a website that can generate a swim set based on a users inputs. this is my flow chartthe user will be inputing their skill level, workout focus and distance goal. im pretty sure my second dataclass is supposed hold like swim_set or something that has the warm up main set and cooldown but idk. I dont want you to tell me exactly how to do it though i need to work thorugh it myselfI think you're actually thinking about your second dataclass in the right way! Your idea that it should represent a SwimSet containing a warm-up, main set, and cooldown makes sense.Let's work through the design without me giving you the finished code.1. First, understand the difference between your two dataclassesThink about your project as having two different kinds of information:Dataclass 1StateThink of this as the website's memory. It keeps track of what is happening in your application.For example:What skill level did the user select?What workout focus did they select?What distance goal did they enter?Has a workout been generated?Dataclass 2SwimSetThink of this as the actual workout that your program creates.For example:Warm-upMain setCooldownThe important distinction is:State stores information about the application, while SwimSet represents the workout your application generates.2. How do they connect?Here's the relationship I would encourage you to think about:USER INPUTSSkill level · Workout focus · Distance goalSTATE DATACLASSStores the user's selectionsWORKOUT GENERATIONUses the inputs to determine the workoutSWIMSET DATACLASSContains the warm-up, main set, and cooldownGENERATED WORKOUT PAGEDisplays the workout to the userThe connection you're looking for is that your State dataclass should have a way to keep track of the generated SwimSet.In other words, you're not combining the two dataclasses into one. You're having one dataclass contain an instance of the other.3. Let's think through your SwimSet dataclassI want you to answer three questions before

writing it.Question 1: If your SwimSet contains a warm-up, main set, and cooldown, what Python data type do you think each of those should be?For example, would each section be:A string?A list of strings?A list of numbers?Something else?Think about how you want to display multiple swimming exercises in each section.Question 2: Should your State dataclass store the actual warm-up, main set, and cooldown individually, or should it store one SwimSet object that contains all three?Think about which approach avoids unnecessary repetition.Question 3: What should happen before the user generates a workout?Your State needs to exist when the website first loads, but the user hasn't generated a workout yet. How might you represent that situation?Hint: Think about Python's None.4. One thing I noticed in your flowchartYour overall page flow makes sense. I particularly like that you separated the workout generation page from the generated workout page.One recommendation: keep your data structure separate from your pages.For example:Your home page collects information.Your State remembers that information.Your workout-generation function creates a SwimSet.Your generated workout page displays that SwimSet.You don't need a separate dataclass for every page.Let's start with Question 1: Do you think warm-up, main set, and cooldown should each be strings or lists? Explain why you think your choice makes sense. I'll help you reason through it without writing the solution for you.well i first thought it could be lists but now im thinking I would need ANOTHER dataclass if i do thatThat's actually a really good observation! You're recognizing that if you want each section to contain multiple swimming exercises, a single string might not be enough.But here's the important thing: You do NOT necessarily need another dataclass just because you're using lists.Let's compare two possible designs.Option A: Lists of stringsImagine a workout looks like this:warmup = ["200 freestyle", "100 kick", "100 pull"]main_set = ["4 x 100 freestyle", "4 x 50 sprint"]cooldown = ["100 easy freestyle"]Each section is simply a list of strings.You wouldn't need another dataclass because Python lists can already hold multiple items.Option B: Another dataclassYou might eventually want each

swimming exercise to have more information.For example:DistanceStrokeNumber of repetitionsRest timeIn that situation, having an additional dataclass representing an individual swimming exercise could make sense.However, that adds complexity, and I'm not sure you need it for your CS1 project.My question for youThink about what you want your generated workout to look like.Would you rather have:A. Simple workout instructions:200 freestyle100 kick4 × 100 freestyleB. More detailed workout information:Distance: 100 yardsStroke: FreestyleRepetitions: 4Rest: 20 secondsI'd lean toward A for your first version.Remember, your goal is to build a working swim-set generator, not create an unnecessarily complicated data structure!yeah for now i think option A is the best. thank you ill work from here and ask more questions if i need any helpYou're welcome! I think Option A is a great starting point for your project.You've got a solid foundation:State = stores user inputs and the generated workout.SwimSet = stores the warm-up, main set, and cooldown.Lists of strings = store the individual swimming instructions.Build it one piece at a time, and don't worry about making everything perfect immediately.Good luck with the project! 😊if I wanted to include a loop somehow, where would be a good place to implement it?I actually think there are a few natural places to implement a loop in your swim-set generator! The best place depends on what you want the loop to accomplish.Here are three possibilities:1. Generating the individual workout sectionsMy recommendationSince you're using lists of strings, you could use a loop to generate or display the exercises in each section.For example, imagine your main set contains:main_set = ["4 x 100 freestyle", "4 x 50 sprint", "200 kick"]A for loop could go through each item and display it on your generated workout page.Conceptually:for exercise in main_set: # Display exerciseThis is probably the easiest and most natural loop to implement.2. Displaying the entire workoutYou could also use loops to display all three sections:Warm-upMain setCooldownThis would let you display multiple exercises without writing separate display instructions for every exercise.3. Calculating the total workout distanceYou could

use a loop to go through the exercises and calculate the total distance of the workout.For example, if your workout contains:200 yards100 yards400 yards100 yardsYour loop could add those distances together.This would be a little more complicated because you would need to extract the distances from your exercise strings.What I would recommendFor your first implementation, I'd use a for loop to display the exercises in your warm-up, main set, and cooldown.Then, if you want to challenge yourself, you could add a loop to calculate the total workout distance.One question to think about: Does your assignment require you to demonstrate a loop as part of your program, or are you just looking for a natural way to incorporate one? That might influence which approach makes the most sense.Final SubmissionWhen you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.These are the project contraints: Final Project Requirements (Specs)Final Project RoadmapThe final project will involve creating a web application with Drafter. The different steps of the project will be spread across the last several weeks of class.Week 10: Brainstorming & Ideation - You will come up with an idea for your web application. You will brainstorm its goal, details, and functionality.Week 11: Design - You will plan how the site will look and how you will implement the requirements listed on this page.Week 12: Development & Deployment - You will develop the code for your website and deploy it.Week 13: User Testing - You will show your website to peers and get feedback on their experiences using it.Week 14: Updates based on Feedback - You will make updates based on the feedback you receive and reflect on the process of creating your web application.Site Functionality RequirementsBelow are the requirements that your site must fulfill. Plan ahead so that meeting these requirements is easier as you develop your project.At least 7 routes.At least 5 Pages, or 10 FragmentsLinks to an external site., or a combination thereof.The implementation of the State dataclass must have at least 4 fields.At least one field must be a list of class instances or a dictionary of class instances.This means that there must

be a second dataclass.At least 4 state fields, including the list or dictionary of class instances, must each be meaningfully modified in at least one route; those 4 state fields cannot be constants.At least 3 input fields (TextBox, CheckBox, SelectBox, TextArea). They don't have to be the same type.At least 3 meaningful if statements.At least 1 loop that iterates over an attribute of the list of class instances or dictionary of class instances in a meaningful way.The site should have a legitimate purpose or functionality.No global variables (global constantsLinks to an external site. are fine).You must use the Drafter library.Any advanced coding techniques (i.e. that weren't covered in this course) used are thoroughly documented in the code.More creative and impressive designs are given flexibility on the above requirements - impress us! However, impressive websites tend to do more than the above requirements. Further, if you miss more than 2 requirements, you will get no credit for that submission. For example, missing a second dataclass means you would be missing not only that requirement, but also the list or dictionary of such dataclass instances, and the loop iterating over that list. This would mean that the submission would get a zero.Code RequirementsMake sure your program and website runs; broken programs do not earn points. This means that if your program contains a syntax error or runs and doesn't deploy a website, you get zero points for this assignment, and are subject to any other penalties listed in the syllabus.Once you are finished with your project code, check over the following:Chosen good variable names (check last two sections of Chapter 1B1Links to an external site.)Unit tested your functions where possible, being valid and thoroughLinks to an external site.: each route must have at least one unit test, if possible (routes that have randomization are difficult to unit test, so they are not required to).If a function can't be unit tested, you must note as much in a docstring or comment.Broke up excessively long lines (80 characters is a good rule of thumb, but we'll be strict about not exceeding 120 characters)Included types in your function headersIncluded good docstrings Links to an external site.as the first line of your

function's bodyDecomposed complicated functions with helper functionsYour TA will be grading your code's quality based on the ideas above.keeping in mind the project contstraints, can you write me a code for this webite